Imitation-Learning-Enabled Vehicular Edge Computing: Toward Online Task Scheduling

Laisen Nie, Xiaojie Wang, Wentao Sun, Yongkang Li, Shengtao Li, and Peng Zhang

Abstract

Vehicular edge computing (VEC) is a promising paradigm to enable information sharing and acquisition among vehicles. Since information freshness has significantly influence on task scheduling, an emerging metric, named the age of information (AoI), has been utilized to evaluate it. Recent research generally focuses on AoI minimization but pays little attention to information personality. However, different impacts caused by distinct information may be posed on user decisions. This article first briefly introduces the state of the art of imitation learning in wireless networks. After that, an imitation-learning-based online task scheduling scheme is designed with the support of VEC. It intends to minimize the average age of critical information (AoCI), referring to the age of information that has significant impacts on vehicle decisions. Performance evaluations show that the proposed scheme outperforms other algorithms from several aspects. At last, we discuss several potential research challenges and open issues for artificial intelligence in the Internet of Vehicles.

Introduction

The Internet of Vehicles (IoV) is a prevalent paradigm to provide content-sharing platforms for users inside high-speed vehicles by vehicle-to-everything (V2X) techniques. Users inside can acquire and share information through the Internet access provided by base stations (BSs) and roadside units (RSUs). To reduce the transmission delay, BSs and RSUs can also integrate with edge servers to provide computation and storage resources for vehicles, which is called vehicular edge computing (VEC) [1]. For instance, RSUs can store updated traffic information and send it to passing vehicles that require traffic information along their routes. As a result, information freshness has become an important metric to evaluate user experience in information-centric IoV systems.

The age of information (AoI) has been proposed to qualify information freshness. Commonly, it is regarded as the elapsed time from the last information update time to now. Its definition was first given in [2] for the purpose of capturing delay requirements of applications that allow periodic information broadcasting. Existing research on AoI is devoted to minimizing the long-term average or peak values constrained on wireless communication resources. However, different user interests, information types, and levels have not been jointly considered, since they have heavy impacts on user decisions.

To realize such a VEC system by considering the above factors is challenging. First, how to characterize user interests, information types, and levels, and evaluate their joint impacts on user decisions are difficult, since no existing model is suitable for such a complicated network environment. Second, the VEC server may not know all user information to do optimal update scheduling, due to the fact that users may be not willing to reveal their private information. Third, for such a partial observation system, a traditional scheduling algorithm is not efficient, and a learning-based algorithm is desirable. However, deep reinforcement learning (DRL) has low efficiency in the initial stage of online training. Thus, we integrate imitation learning with our VEC system, taking its advantage of fast convergence speed and near-optimal performance. To the best of our knowledge, we are the first to apply imitation learning for AoI-based scheduling in VEC networks.

This article proposes an imitation-learning-based online scheduling approach to minimize the average age of critical information (AoCI) of vehicles in IoV systems. AoCI is the age of information that has impacts on user decisions. We first introduce the state of the art of imitation learning. Then we establish a VEC-enabled network framework. Following that, an imitation-learning-based information update scheduling approach is proposed. To obtain the expert policy, we design a heuristic searching algorithm based on full network states, which can be executed offline. Then we train the learning model online to enable the learning agent to mimic the behaviors of experts. Experimental results show that the designed algorithm can reach near-optimal performance compared to that of the expert. Finally, we discuss several research challenges and open issues for artificial intelligence (AI) in IoVs.

The State of the Art of Imitation Learning

The learning agent in imitation learning can mimic expert behaviors. Through training on expert demonstrations, the relationship between states and actions can be learned by mapping. In the

Laisen Nie, Wentao Sun, and Yongkang Li are with Northwestern Polytechnical University; Xiaojie Wang (corresponding author) is with Chongqing University of Posts and Telecommunications; Shengtao Li is with Shandong Normal University; Peng Zhang is with the China Aerospace Science and Technology Corporation.

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following, four promising aspects of imitation learning are introduced.

Behavioral Cloning

Behavior cloning refers to learning from demonstrations provided by experts with the purpose of imitating their behaviors [3]. Specifically, states and actions should be provided with the best choice, regarded as an expert policy. Basically, imitation learning can be treated as a supervised learning problem [4]. To overcome the large generalization error caused by behavioral cloning, generative adversarial networks have been utilized to model the distribution of expert actions. A generative model can be trained, and a discriminative classifier can be utilized to distinguish the data generated by the generative model and the expert policy [5].

Learning from Suboptimal Demonstrations

Although behavior cloning is an efficient method to perform policies based on expert demonstrations, there may not always be demonstrations with good performance. Therefore, researchers seek to find good policies that can learn from suboptimal demonstrations. The authors in [6] introduce a novel learning algorithm that can infer high-quality reward functions based on a group of poorly performed demonstrations ranked by a potential metric. However, the required ranked demonstrations are difficult to obtain in reality, restricting its application for real-world scenarios. Unlabeled suboptimal demonstrations are leveraged in [7], and multiple methods with different qualities are modeled as correlated Gaussian processes. An optimization policy is proposed to recover the correlations between the policies adopted by experts and those by the learning agent. Nevertheless, a majority of expert demonstrations are required, which is not data-efficient.

Learning from Observations

Another kind of imitation learning is to learn from observations where the actions taken by experts are unknown. For example, the learning agent can learn from videos without actions [8]. This is similar to the imitation behaviors of humans and animals, which observe another individual's behaviors and figure out what action leads to his/her behaviors. To reduce the required environment samples for policy learning, a novel approach that can infer latent policies directly from observations is proposed in [9]. The causal effects of latent actions for observations can be characterized, and their likelihood can be estimated. Then the mapping between latent actions and real-world ones can be determined by some environment interactions. After that, the actions on observations can be inferred, and a whole demonstration containing states and actions can be rebuilt.

Integration with Deep Reinforcement Learning

DRL has achieved significant success in decision making problems. Nevertheless, DRL-based algorithms need bulks of data before achieving good performance, limiting the applications of DRL for many real-world tasks. Thus, imitation learning can be integrated with DRL to accelerate the learning rate from a small set of demonstrations [9]. In addition, inverse reinforcement learning can be utilized to deduce a reasonable reward function for DRL from demonstrations, which can explain the expert behaviors and guide the learning agent to take proper actions [10]. In addition, the balance between exploration and exploitation is a key issue for DRL. Imitation learning can be utilized to overcome difficulties caused by exploration. Even multiple-step behaviors of robotics can be controlled by learning demonstrations [11].

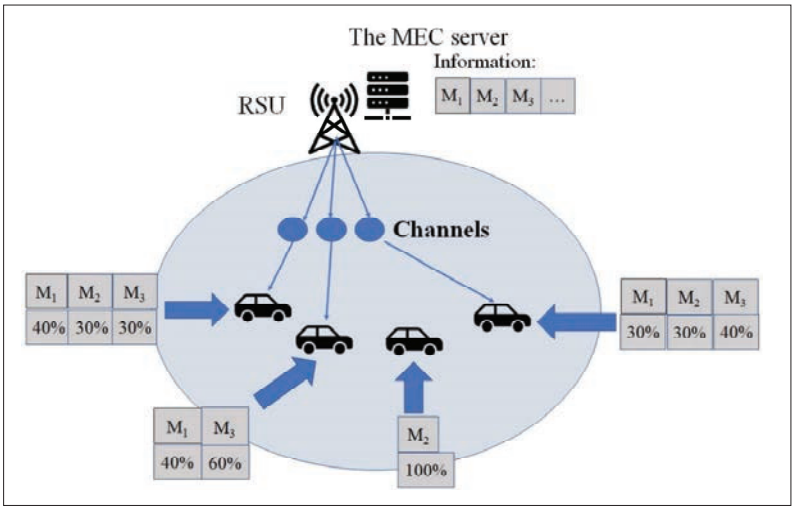


FIGURE 1. Illustrative system model.

Imitation-Learning-Based Scheduling in IoV

In this section, we first present the system model. After that, an imitation-learning-based information update scheduling algorithm is proposed to resolve the AoCI minimization issue in IoV systems. At last, system performance and results are provided.

System Model

As shown in Fig.1, we consider that there are RSUs deployed along roads, and each RSU is connected to an VEC server. When a vehicle moves into the wireless communication coverage of an RSU, it can send information update requests to the RSU, and the RSU can forward these requests to the connected VEC server for information update. For simplicity and without loss of generally, we consider that the information in the system can be categorized into several kinds, and each one has certain levels. For example, there are three kinds of information in Fig. 1, that is, M_1, M_2, and M_3, and each vehicle has different interested ratios for the three kinds of information. For instance, the second vehicle counted from right to left is 100 percent interested in M_2. Due to the resource constraint, the maximum number of vehicles that the VEC server can serve in one time slot is bounded. With the purpose of modeling the impacts of information, we define AoCI as the age of critical information that has impacts on users' decisions by jointly considering various user interests, information kinds, and levels. Our objective is to minimize the AoCI of vehicles, implying the capture of the level change of information as early as possible to avoid vehicles making false decisions.

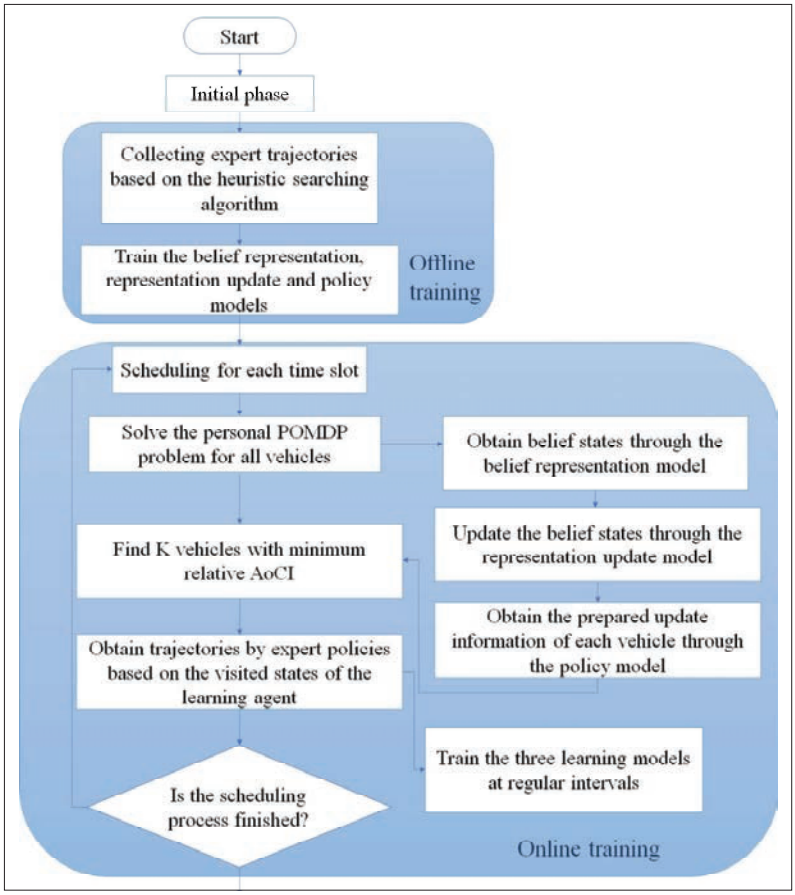


FIGURE 2. The flow chart of the designed learning architecture.

When one kind of information for a vehicle is updated in the current time slot, its AoI decreases to 1; otherwise, its AoI increases by 1. The interested ratio of one vehicle in a kind of information can be defined by a decimal number between 0 and 1, and the total ratio for all kinds of information of a vehicle equals 1. Specifically, we can formulate the AoCI of one information kind for one vehicle by the product of the AoI and the impact of the level change for that information kind from time slots $t-1$ to t. The impact of the level change for one kind of information can be formulated as an exponential growth with the difference between the levels in both time slots $t-1$ and t. Thus, the AoCI for one vehicle can be obtained based on all its interested information. Overall, our objective is to minimize the average AoCI for all vehicles from a long-term perspective.

This optimization problem is constrained by the maximum number of vehicles connected to the VEC server, the channel capacity, total information kinds, and levels. Generally, DRL is applied to solve the formulated problem. However, it always has low convergence speed and poor performance at the first 1000 algorithm iterations, which is not acceptable for online scheduling. To conquer the above disadvantages, we leverage imitation learning, which allows the agent to mimic the behaviors of experts and has good performance from the system's start.

An Imitation-Learning-Based Scheduling Algorithm

Problem Transformation: For simplicity and without loss of generality, we consider that there is one VEC server along with some vehicles in its coverage, which can be expanded to the scenario of multiple VEC servers. Our objective is reduced to minimizing the average AoCI of vehicles in the coverage of one VEC server in each time slot. For this subproblem, we merely need to find K vehicles with the minimum relative AoCI in each time slot. Consequently, we can conduct a heuristic searching process including two steps. In the first step, the information that can reach the minimum AoCI of each vehicle through the update process should be identified based on the channel capacity and its size. A dynamic programming algorithm can be conducted to find the possible candidate information for update. In the second step, K vehicles with the minimum AoCI computed by the first step can be selected. A greedy algorithm is leveraged to find the K vehicles with the minimum AoCI.

Partially Observable Markov Decision Process (POMDP): Although the above-mentioned heuristic searching process can derive the optimal solution, it requires full knowledge of vehicles' local information. This is impossible in reality, since individuals do not intend to expose their private information to the VEC server. Thus, the heuristic searching process is not suitable for online learning. Fortunately, we can collect the optimal solution offline by recruiting volunteers that are willing to expose their information and form expert trajectories. For online learning based on partial observations, we can map our scheduling issue to a POMDP setting. Formally, a discrete-time finite horizon POMDP can be represented by seven elements, that is, the state space, the action space, observations, the reward function, state transition probabilities, conditional observation probabilities, and the time horizon. Since the learning agent cannot directly observe the environment, only an observation can be received, while the real state is hidden. The environment transmits from one state to another based on state transition probabilities, and the received observation can be obtained according to conditional observation probabilities.

In our algorithm, the real state at time slot t includes interested information, interested ratios, and information levels of each vehicle. The focus is on the selected vehicles and their prepared update information. However, the state and action spaces are rather large, generally resulting in huge learning delay, and are unacceptable for online learning. Consequently, we consider a single POMDP setting for each vehicle to seek its candidate information for update, which is the most important process of our AoCI minimization problem. For the single POMDP setting, the real state of one vehicle at time slot t includes the interested information identities, the interested ratio of each information, current information levels, and the time duration for time slot t. The observation contains partial elements in the real state, whereas the local information levels and interested ratios are not in the observation.

Imitate Expert Demonstrations: The learning agent in imitation learning can mimic expert behaviors from demonstrations, which means to train the agent policy based on the dataset of expert trajectories. Therefore, we first train a learning model offline and then refine this model through online interactions with the environment.

To obtain good demonstrations, we assume that some volunteers willing to reveal their private information can be recruited to be involved in the data collection processes through heuristic searching. Herein, the expert policy can be defined as a mapping from the real state to the action based on the optimal searching process described in the last paragraph. Then a dataset containing real states, actions, observations, and rewards can be obtained.

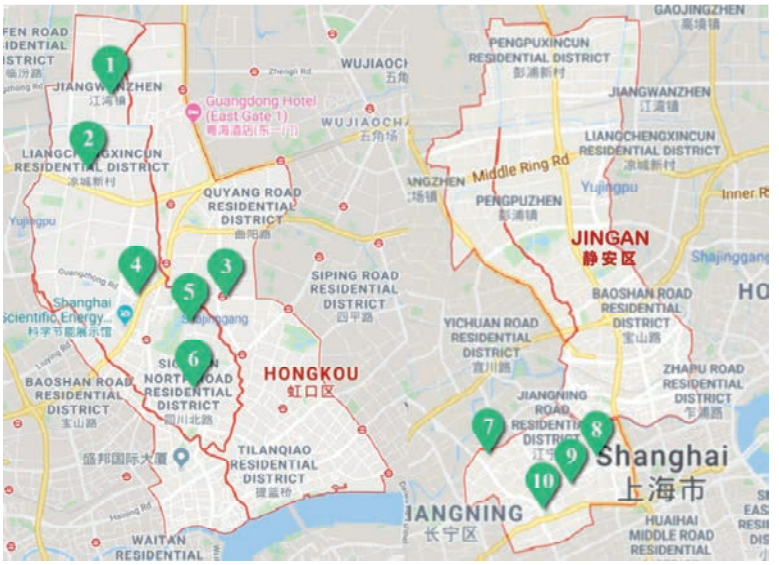
Policy Training: After obtaining the expert dataset offline, we can train the agent policy. Since vehicles are not willing to reveal their private information in an online manner, we should train the policy that can map the observation to the proper action. It is difficult to train a policy that directly maps the partial observations to the proper action. Thus, we should first find the relationship between the observations and the real states, and then map the real states to the proper actions. Three training models are contained in this process: belief representation model, representation update model, and policy model.

The belief representation model enables an inference between the observation and the real state. We input the observation, and the model outputs the belief state, which represents the inferred real state. We train a generation model based on expert trajectories including observations and real states. We consider that a priority distribution coming from parametric families of the distribution generates state s_t, and a likelihood from possibility distribution related to states and observations generates observation o_t. Thus, the purpose of this model is to train the policy model based on current state, observations, actions, and rewards that can approach the true posterior density by minimizing the gap between the estimated distribution and the true distribution (i.e., the Kullback-Leibler divergence).

For online learning, we need to utilize the observation, the action, and the reward to update the belief state. However, the observation, the action, and the reward cannot be obtained beforehand. Thus, a representation update model is necessary to update the belief state based on history observations, actions, and rewards. The training data can be obtained from expert trajectories, and the model can be trained offline. Following that, the representation update model can be learned.

After that, we can train the policy model. Since the belief state can be obtained by the belief representation model and the representation update model, the mapping between the belief state and the action can be constructed.

Overall Steps: In our designed scheduling algorithm, we conduct the algorithm processes as follows. We first collect expert data by executing heuristic offline searching to form expert demonstrations. Then, for online scheduling, we map the scheduling issue into POMDP settings. That is, we train models that can infer relationships between observations and proper actions that is, q_φ, ω_σ, and π_ε are trained based on datasets extracted from expert trajectories beforehand, as shown in Fig. 2. For online scheduling, there are total N POMDP problems in each time slot, where N is the number of vehicles in the corresponding time slot. For solving each POMDP problem, we try to find appropriate scheduling sequences of local interested information for the corresponding vehicle, with the purpose of minimizing personal AoCI. Since the learning agent cannot obtain the interested ratios and levels of local information for vehicles, the training models are leveraged here to find proper actions for each vehicle based on current observations. After that, the possibilities for information update related to each vehicle can be obtained, and we can select the information with large possibilities as the candidate update information. At last, K vehicles with the minimum personal AoCI will be selected.



	Hongkou district		Jingan district
ID	Locations	ID	Locations
1	31.304725,121.473037	7	31.234170,121.431700
2	31.294622, 121.470535	8	31.230982, 121.456374
3	31.276524, 121.491138	9	31.228587,121.455745
4	31.276615, 121.477732	10	31.223078,121.446153
5	31.273372, 121.487506		
6	31.263758, 121.486227		

FIGURE 3. Selected GPS in Shanghai.

Performance Evaluation

We utilize Python and TensorFlow in our simulation to validate the performance of the designed imitation-learning-based scheduling algorithm, named LISA. We consider a dataset containing real-world taxi trajectories in Shanghai from April 1, 2015 to April 30, 2015, in which more than 1000 taxies are recorded [12]. We select some locations in two districts in Shanghai as the locations of VEC servers for experiments, as shown in Fig.3, since the traffic densities of those locations in the two districts are quite different with denser traffic in Hongkou. Moving vehicles can send update requests to those servers within the communication range of those servers. The communication coverage of each VEC server is 200 m. The total number of information kinds is 10,

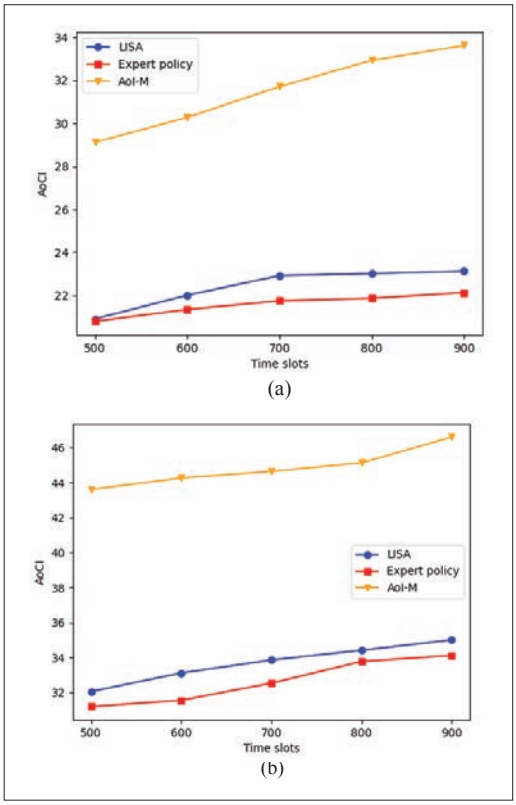


FIGURE 4. Average AoCI with different number of time slots: a) Jingan district; b) Hongkou district.

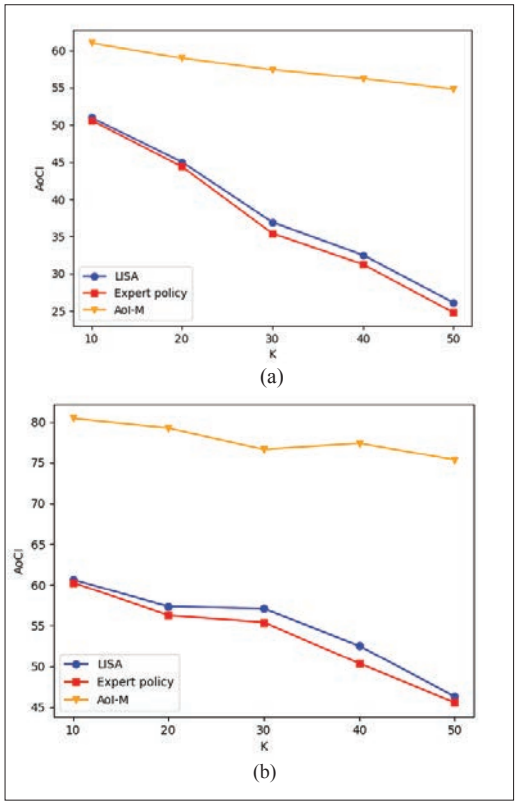


FIGURE 5. Average AoCI with different number of *K*: a) Jingan district; b) Hongkou district.

while that of information levels is 5. We randomly set the interest information and its update for each vehicle in each time slot. The information size is between 1 MB and 5 MB, and the wireless transmission power between the VEC server and the vehicle is 10 dBm with noise power −172 dBm. We collect data for 200 time slots offline to train expert policies. We leverage four convolutional layers and two fully connected layers with ReLU nonlinearities for the belief representation model, and four fully connected layers for the belief update model and policy model, respectively. The Adam optimizer is utilized to train the three models.

The following two algorithms are compared.

Expert Policy: Heuristic searching is utilized to schedule information update for vehicles, and the learning agent is assumed to be aware of all the knowledge of vehicles.

AoI-M [13]: It is an information update scheduling algorithm aimed at minimizing the AoI of vehicles. Varying sample sizes, multiple transmission units, and different sample behaviors are considered in this algorithm.

We evaluate the average AoCI in the considered two districts, Jingan and Hongkou. Figure 4a shows the trend of average AoCI with different numbers of time slots in Jingan district. We can observe that the expert policy is always the best. The performance of LISA is near-optimal, and that of AoI-M is the worst. This is because based on the awareness of vehicles' local information, the expert can find the best policy for the considered AoI minimization issue. LISA is an imitation-learning-based scheduling algorithm, which can mimic the expert behaviors. However, it cannot perform better than the expert. AoI-M aims to minimize the average AoI by randomly selecting information to update. Figure 4b shows the trend of the average AoCI with different numbers of time slots in Hongkou district. We can see that the performance of the three algorithms is worse than that in Jingan district. The reason is that the number of vehicles in Hongkou district is larger than that in Jingan district, and the information of vehicles in Hongkou cannot be updated in a timely manner.

We also evaluate the average AoCI with different values of *K* for all three algorithms, where *K* is the maximum number of vehicles that can be connected to the VEC server simultaneously in one time slot. Figure 5a is the trend in Jingan. When the value of *K* becomes large, the average AoCI of the three algorithms becomes small. The reason is that when the value of *K* becomes large, more vehicles are allowed to update information simultaneously. For the comparison of those three algorithms, the expert policy is always the best, while our designed imitation-learning-based algorithm is close to the expert, while the performance of AoI-M is far from those of the others. The reason is similar with that of Fig. 4. Figure 5b is the trend of the average AoCI in Hongkou. With more vehicles in Hongkou, the average AoCI is higher than that in Jingan. That is to say, more vehicles need to update their local information, and they may have a long waiting time for updating.

Figure 6 illustrates the loss trends of the agent policies based on the dataset of Jingan. We can observe that the loss is very small even at the beginning of the iterations. This is because our

imitation-learning-based scheduling algorithm allows the agent to imitate expert behaviors, and the agent can quickly learn the corresponding policies based on the provided observation-action pairs encountered by the expert.

Research Challenges and Open Issues

In this section, we discuss several research challenges and open issues for the application of AI technologies in IoV systems.

Large State and Action Spaces

Although AI technologies have achieved great progress in robotics and online games, there are still many obstacles for the applications in IoV systems. This is because IoV systems have many vehicles with distinct purposes, speeds, and directions. To satisfy distinct vehicular requirements, the coordinate mechanism should be designed based on all vehicles, which may result in large state and action spaces for the learning-based algorithm. Nevertheless, most existing DRL-based algorithms are inefficient in handling such large state and action spaces. One possible solution is to simplify the considered problem, which can potentially reduce the sizes of states and actions. Another solution is to design effective AI solutions that can handle the large state and action spaces.

Learning Model Offloading

Currently, only a few applications leverage AI technologies (e.g., deep learning) on mobile terminals, since they consume lots of computation resources. Due to the insufficient infrastructure support, researchers investigate algorithms to offload the learning model from local devices to edge servers. However, offloading the computation-intensive learning model, especially the involved neural network to the edge server, faces many challenges under the requirements of latency, communication efficiency, and available network resources. The first is how to treat the learning model, for example, treating it as an entity or partitioning it into several portions. The second is which kind of offloading models should be selected, for example, partial offloading or entire offloading. The third is that when a series of clouds, edge servers, and terminals exist, where to offload the generated tasks of vehicles should be designed carefully to meet the requirements of vehicles.

Performance Efficiency

AI-based technologies, especially DRL, are always based on historical experiences to make decisions on strategy selection. Therefore, they require a long time to learn the optimal selections for highly dynamic networks. In other words, the learning algorithm needs a long time to converge, resulting in poor system performance at the initial stage. Thus, imitation learning is promising to avoid the poor performance brought by DRL at the beginning stage of learning. In addition, for the multiple-objective system, how to set the reward function to obtain a balanced result is necessary to investigate in IoV systems, since there are always trade-offs between the required resources and the performance efficiency.

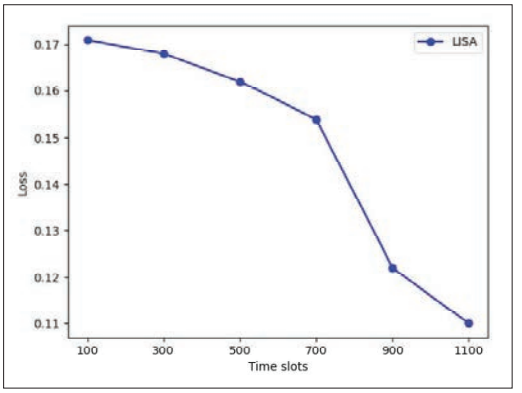


FIGURE 6. Performance of losses for agent policies.

Unstable Topology Handling

Due to high speeds and unpredictable movements of vehicles, the topology in IoV systems is unstable and changes with time [14]. Traditionally, AI-based technology is capable for issues in stable environments, and the agent can learn the behaviors of participants. However, in IoV systems with dynamic vehicle flows, the learning agent may always have unfamiliar environment elements, making its decision making inefficient to cope with the requirements of unencountered vehicles. Thus, how to design a learning-based algorithm that can handle the dynamics of the network topology in IoV deserves to be studied. In addition, how to learn the network dynamics from both temporal and spatial dimensionalities is another research direction for traffic flow prediction in IoV.

Partial Observations

Generally, IoV systems are built based on vehicular ad hoc networks, where vehicles communicate with each other in an opportunistic manner. Thus, they do not have full system observations, and cannot always make the optimal decision for communication, computing, and caching [15]. Existing heuristic scheduling algorithms are not efficient for such a partial observation environment, calling for novel algorithms. Although AI-based technology is efficient for learning in an unknown environment, applying it in IoV systems faces the challenges of dynamic network topologies, vehicle quantities, and mobility. Another issue is where and how to conduct the learning algorithm under partial observations to achieve good performance.

Multi-Client Learning

For vehicles moving on the same road segment or toward the same destination, they can cooperate for a common purpose, such as information forwarding, or struggle for their own purposes. Although game theoretical models can reach a Nash equilibrium in many existing bargain environments, they cannot be directly applied in such ad hoc networks without a centralized server, since global information is lacking. Thus, how to make vehicles reach a consensus and guarantee fairness among them needs to be investigated. In addition, when and how to exchange local information related to learning algorithms between two neighboring vehicles are important to improve the performance of the learning algorithm.

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Conclusion

Information freshness is an important metric in information-centric networks, such as traffic management in IoV. By considering the importance of information with different impacts on user decisions, this article has proposed an imitation-learning-based information update scheduling algorithm in IoV. It aims to minimize the average AoCI for vehicles by solving the formulated optimization problem. We first give a brief introduction of imitation learning, and then specify how to leverage imitation learning in the considered IoV systems. Performance results demonstrate the effectiveness of our designed algorithm from the aspects of the average AoCI with different metrics. Finally, we discuss several research challenges and open issues, focusing on the issues of AI-enabled technologies in IoV systems.

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Biographies

Laisen Nie received his Ph.D. degree in communication and information systems from Northeastern University, Shenyang, China, in 2016. He is an associate professor in the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include network measurement, network security, cognitive networks, mobile edge computing, and machine learning.

Xiaojie Wang received her M.S. degree from Northeastern University in 2011 and her Ph.D. degree from Dalian University of Technology, China, in 2019. From 2011 to 2015, she was a software engineer with NeuSoft Corporation, China. Currently, she is a distinguished professor at Chongqing University of Posts and Telecommunications, China. Her research interests are wireless networks, mobile edge computing, and machine learning.

Wentao Sun received his B.Eng. degree in computer science and technology from Shandong Normal University, Jinan, China, in 2020. And he is a graduate student in the School of Electronics and Information, Northwestern Polytechnical University. His research interests include network measurement, anomaly detection, and link prediction.

Yongkang Li received his B.Eng. degree in electronic information engineering and his Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2011 and 2016, respectively. From August 2015 to August 2016, he was a visiting Ph.D. student with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. Since January 2017, he has been an assistant professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include radar signal processing and network security.

Shengtao Li received his M.S. degree in operational research and cybernetics from Ludong University, China, in 2010, and his Ph.D. degree in control theory and control engineering from Northeastern University in 2013. He is currently an associate professor in the School of Information Science and Engineering, Shandong Normal University. His research interests include nonlinear systems theory, optimal switch-time control theory of switched stochastic systems, and robust control of systems with time delay.

Peng Zhang received his M.E. degree from Northeastern University in 2013. From 2013 to 2020, he was an engineer with Aerospace Star Technology Corporation, China. Currently, he is a supervisor at the China Aerospace Science and Technology Corporation, China. His research interests are satellite communications and capital operation.